

AMEL MUTHU

Enterprise AI & Generative AI Engineer

Azure AI Apps and Agents | Agentic AI | RAG | Databricks

Dubai, UAE | +971 55 113 9510 | amelmuthu@outlook.com | github.com/amalmuthu

PROFESSIONAL SUMMARY

Enterprise AI & Generative AI Engineer with 5+ years of experience across Data & AI, including 3+ years focused on AI/ML and Generative AI solutions. Experienced in designing and developing enterprise AI applications using Azure AI, Agentic AI, RAG, Databricks, Semantic Kernel, vector search and applied machine learning. Hands-on experience integrating AI solutions with enterprise data, APIs, databases and business workflows, with a focus on secure, scalable and production-oriented implementation.

CORE EXPERTISE

Generative & Agentic AI: Azure OpenAI, Azure AI Foundry, LLM applications, RAG, prompt engineering, Ollama, Qwen.

Retrieval & Document Intelligence: Azure AI Search, FAISS, pgvector, Embeddings, Hybrid & Semantic Search, Reranking, Sentence Transformers, Contextual Retrieval, OCR, PaddleOCR, Policy & Document Analysis.

AI Agents & Orchestration: CrewAI, Semantic Kernel, Azure AI Agents, LangChain, LlamaIndex, Multi-Agent Workflows.

AI Tool Integration: Model Context Protocol (MCP), Function/Tool Calling.

AI / Machine Learning: Python, Pandas, NumPy, Scikit-learn, TensorFlow, Keras, PyTorch, NLP, predictive modelling, feature engineering, model evaluation.

Data & Cloud Engineering: Azure Databricks, SQL, REST APIs, FastAPI, Azure App Service, Oracle, MySQL, Microsoft SQL Server, data transformation and analytics.

Analytics & Visualisation: Power BI, Tableau, Matplotlib, Seaborn, statistical analysis, KPI reporting.

PROFESSIONAL EXPERIENCE

Data Scientist / AI Engineer | Iterative Research Tech Limited | Nov 2023 - Present

- **Azure Multi-Agent Content Platform:** Architected a sequential multi-agent content-generation platform using Azure OpenAI, Azure AI Agents, Azure AI Search, Databricks, Fast API, and MCP, orchestrating specialized agents across content-generation and publishing workflows.
- **AI-Powered Procurement Intelligence:** Designed an enterprise procurement platform using Qwen3:8B, Ollama, RAG, pgvector, PostgreSQL, and Python, supporting policy-grounded SOW/RFQ generation, budget benchmarking, vendor evaluation, and procurement decision support. The solution keeps procurement data on-premise and uses pg vector-backed policy retrieval.
- **Explainable Vendor Decision Engine:** Engineered deterministic vendor scoring across compliance, price, delivery, certifications, and vendor history, with traceable data lineage and LLM-generated business explanations and recommendations.
- **GRC & Compliance Intelligence:** Developed a multi-framework platform for UAE IA, PDPL, and ISO 27001, combining semantic control mapping, RAG-based document assessment, compliance tracking, and executive reporting.
- **Document Intelligence & Semantic Retrieval:** Built OCR/NLP pipelines using FAISS, Sentence Transformers, contextual retrieval, and Paddle OCR for semantic document search and policy analysis.
- **Conversational Enterprise Data Access:** Developed an adaptive-learning AI application translating natural-language questions into database queries, enabling non-technical employees to retrieve enterprise information without SQL knowledge; placed 3rd in an eight-day AI challenge conducted by DDS UAE.
- **Cryptocurrency Analytics MCP Server:** Developed a FastMCP 2.0 server connecting Claude with PostgreSQL/TimescaleDB, exposing 8+ specialized tools across 1,935 funding-rate tables for conversational analytics.

- **Multilingual Voice & TTS:** Built an open-source desktop application integrating LuxTTS, Qwen3 TTS, and MMS5 for multilingual TTS and voice cloning, with MCP-based tool integration for agent-driven voice workflows.
- **Healthcare Predictive Analytics:** Developed and evaluated a post-PCI mortality-risk model using 8,296 patient records, identifying mechanical support devices and severe anaemia as leading risk indicators.
- **Multilingual NLP:** Developed an English-Arabic job-description optimization and AI-assisted translation application using Streamlit and Hugging Face Transformers.

Data Analyst | Net Solutions | 2012 - 2013

- Analysed operational datasets to identify trends, outliers and improvement opportunities; validated and transformed raw data to improve data quality by 30%.
- Developed interactive Tableau dashboards that reduced report-generation time by 40% and supported KPI-based decision-making.
- Applied statistical analysis to support product and operational improvement strategies.

Database Administrator | Cognizant Technology Solutions | 2010 - 2011

- Designed data models from remotely sourced XML files and supported DataStage jobs, performance tuning and production data troubleshooting.
- Analysed metadata dependencies, prepared job sequences and produced system specifications and ETL source-to-target mappings.

CERTIFICATIONS

- Microsoft Certified: Azure AI Apps and Agents Developer Associate
- Microsoft Certified: Azure Databricks Data engineer Associate
- Google Data Analytics Professional Certificate
- Oracle Certified Professional (OCP), Oracle Database 11g Administrator
- Oracle Certified Associate (OCA), Oracle Database 11g Administrator

EDUCATION

Bachelor of Engineering in Computer Science | Anna University, Chennai